

How do you slice a waffle?

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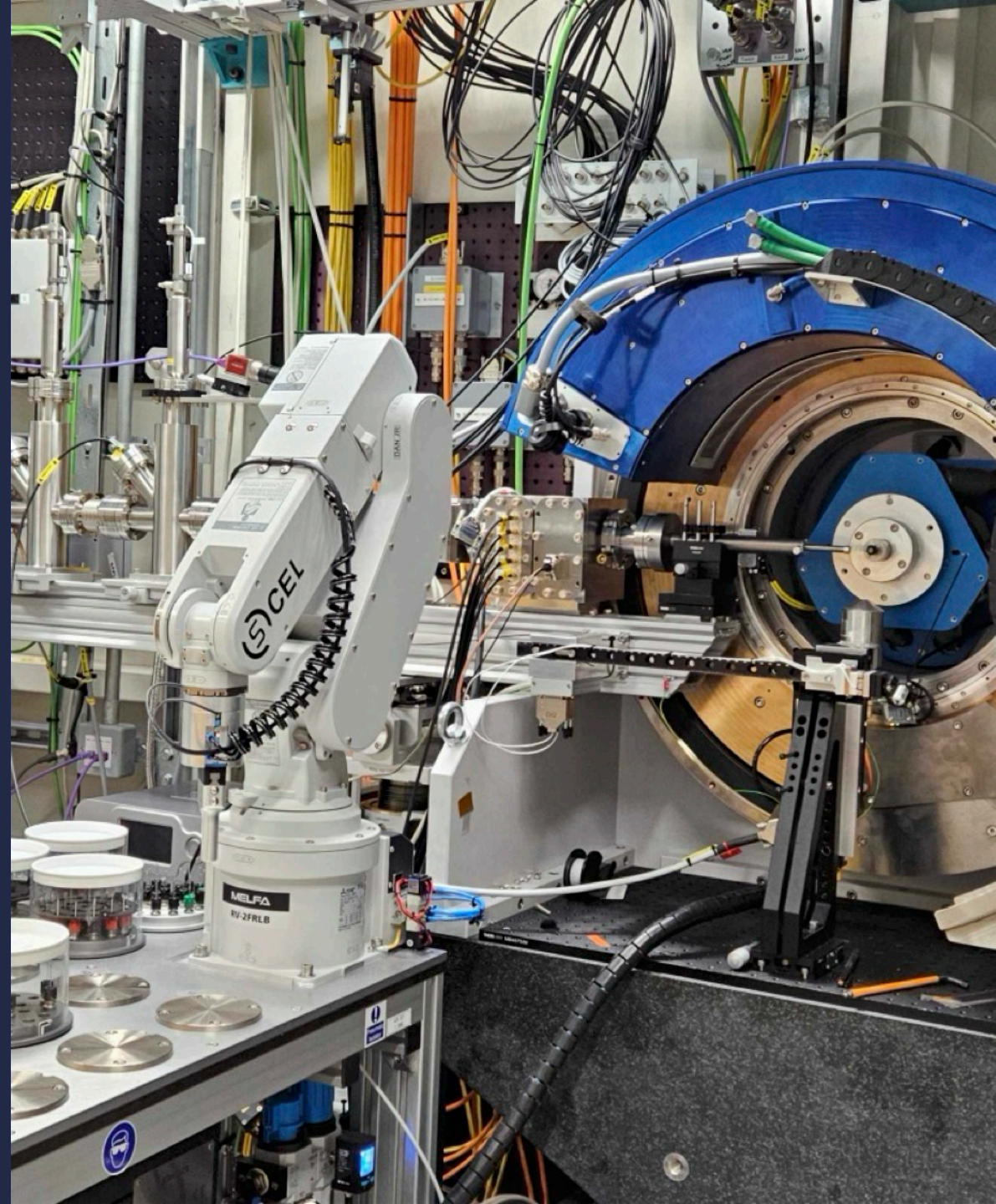
20th October 2025

Contents

- a quick introduction to how i15-1 operates
- beamline schedule for next 12 months
- what project waffle *hoped* to achieve
- establish a shortlist of well-defined functional statements

A quick intro to I15-1

We specialise in *total scattering* which most people inspect via the *pair distribution function*



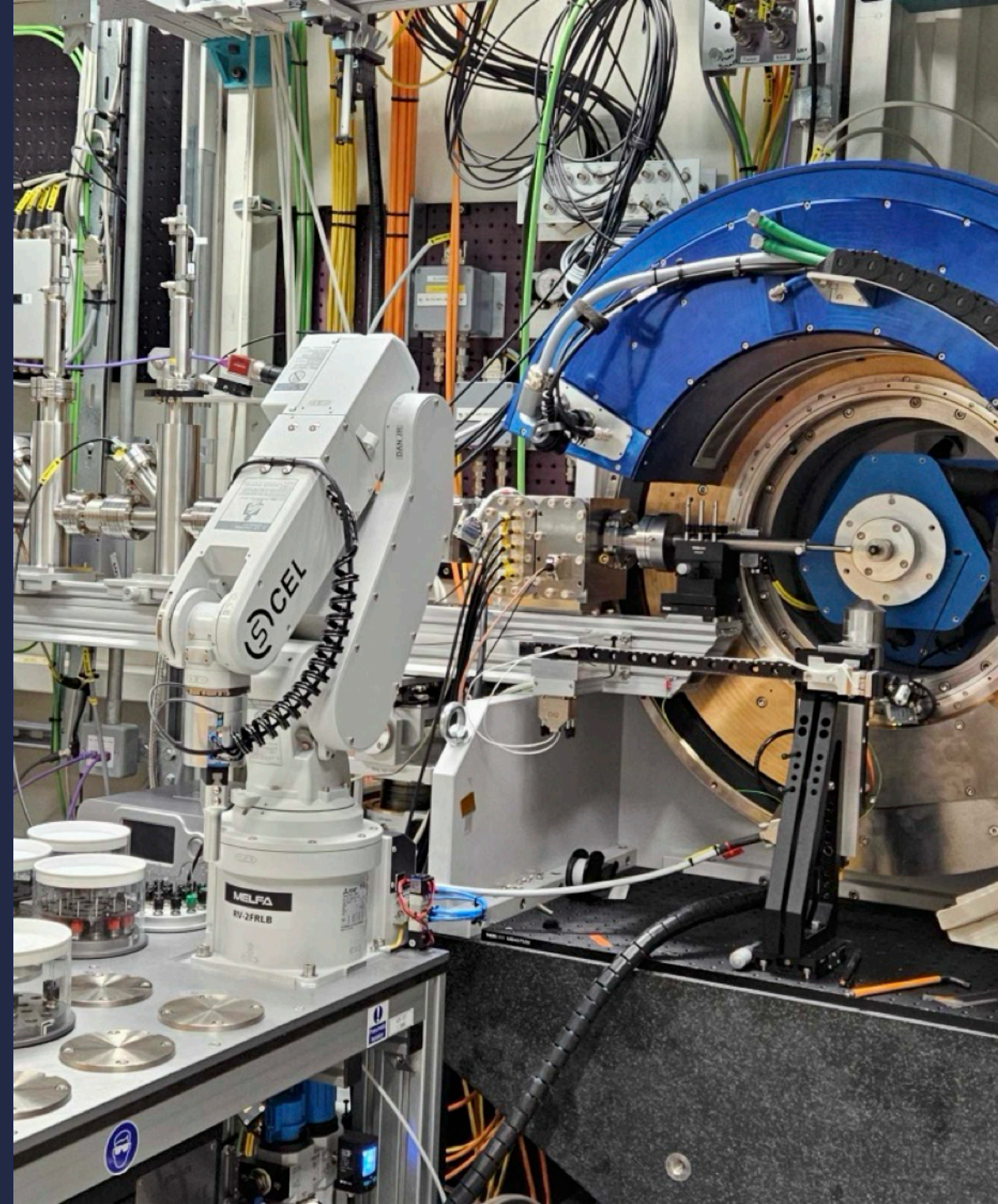
Our detector is a Perkin Elmer (sometimes called a Varex) which operates on a fixed clock: it reads out every 3 seconds

- we have a new detector though 😊
- we don't use it because the data it produces isn't good enough 😞

Types of experiment

Roughly half of proposals study sample *ex situ*; we use a robot sample changer

Some of those we run as mail-in



Calculating the Pair Distribution Function is *not* straightforward

- we need to subtract the background
- we need to know the sample composition and atomic density

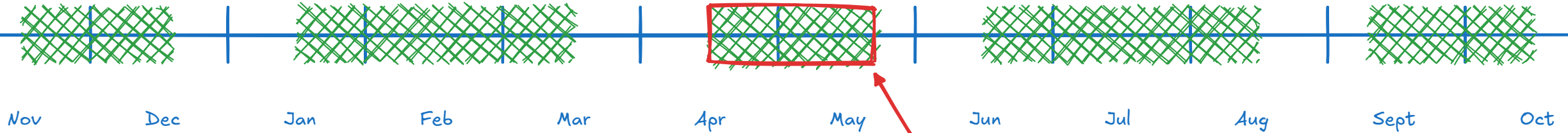
get 6 new
ARC modules

get 15 new
ARC modules

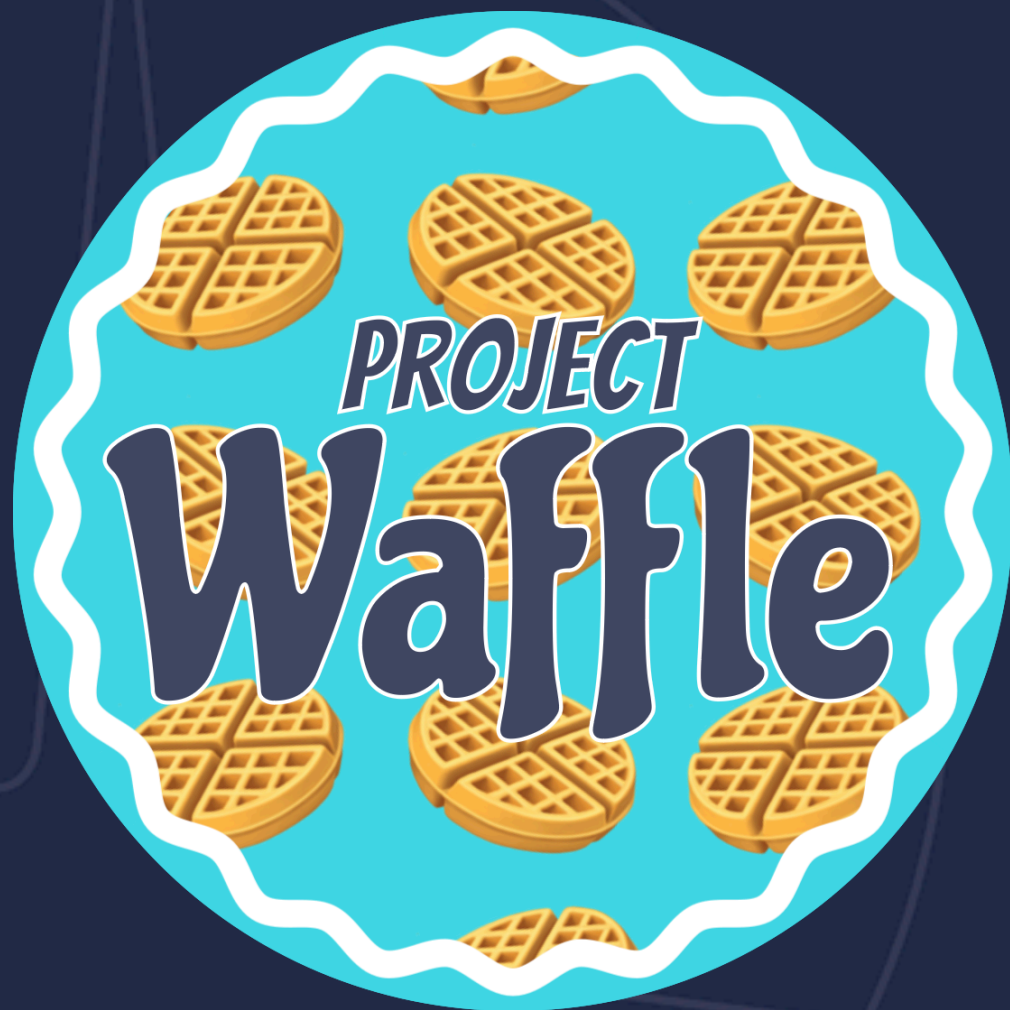
AP38

AP39

AP40

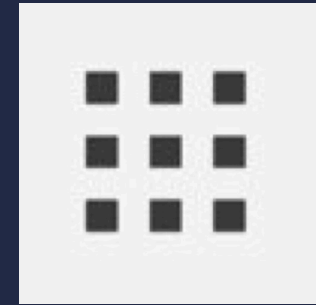


Motion controller upgrade



what is... *was*...
project waffle?

Waffle



[**wof**-uhl]

noun

In user interface design, "waffle" refers to a grid-like menu or app launcher, often used as a navigation element, resembling a waffle's squares.

Project waffle ~~seeks~~ *sought* to bring together new software components to make a fully integrated system on an existing Diamond beamline (I15-1).

beamline objectives

A measurement takes 10 minutes of beamtime, and an hour of staff-time

- collecting sample information
- distributing consumables
- making experiment plan
- processing data
- notifying users

WAYS TO SLICE A BREAKFAST WAFFLE

